Notes on lecture 11

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1 Volume of solid of revolution

Let us start with volumes of some known shapes. The volume of a circular cylinder equals $\pi r^2 h$, where h is the height of the cylinder and r is the radius of the base of the cylinder. Please see Figure 1.0.1 below.

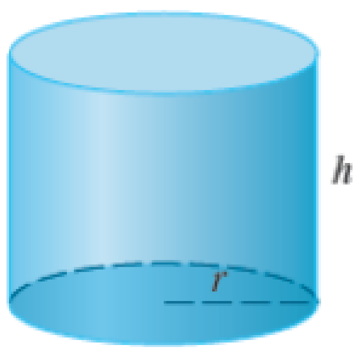


Figure 1.0.1: A circular cylinder

Likewise, if we have a rectangular box with breadth, length, and height equal to x, y, and z respectively. Then its volume is xyz. Please see Figure 1.0.2 below.
These were some known shapes. Now let us work out and see how to find the volume of a randomly shaped solid.
Consider a solid S as shown in Figure 1.0.3 below, which lies within the boundaries of the interval $[a, b]$ along the x-axis. The solid shown in Figure 1.0.3 lies in the xyz-space, not in the xy-plane.
At a distance x from the origin (along the x-axis), let us consider a plane

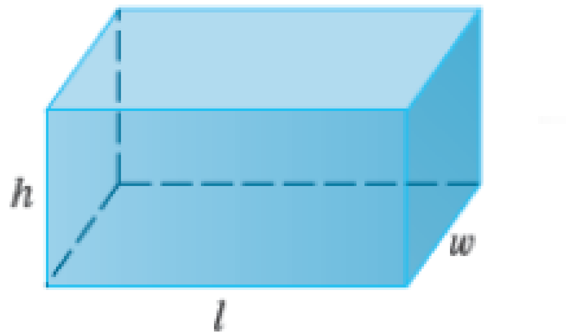


Figure 1.0.2: A rectangular box

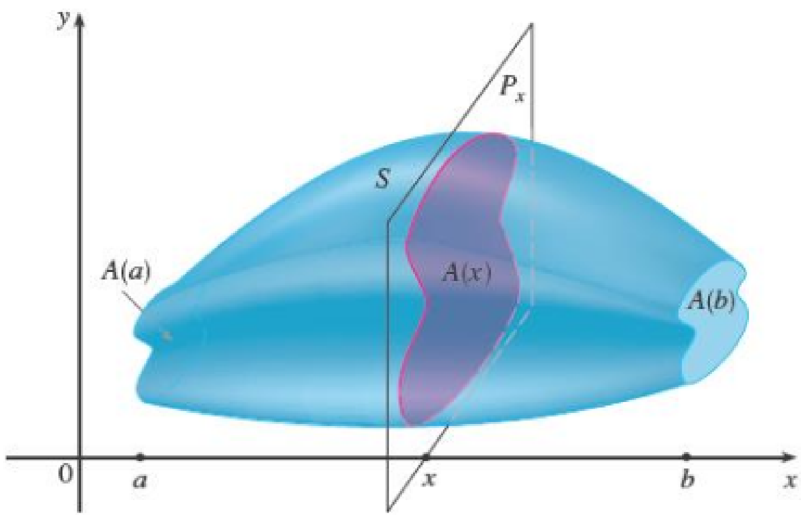


Figure 1.0.3: A solid being cut by a plane P_x

P_x perpendicular to the x-axis. This plane P_x will cut the solid as shown in Figure 1.0.3. Let $A(x)$ denote the cross sectional area of the cut of the solid. Let us now divide the interval $[a, b]$ into n subintervals $[x_{i-1}, x_i]$, where $1 \leq i \leq n$. The intervals need not be equal in length. But we may assume it for the sake of simplicity. Let x_i^* be a point in the interval $[x_{i-1}, x_i]$. Corresponding to the end points of the subinterval $[x_{i-1}, x_i]$, we have 2 planes

$P_{x_{i-1}}$ and P_{x_i}. If we cut the solid by these 2 planes, we get a slice of the solid. See Figure 1.0.4 for an illustration. Let $A(x_i^*)$ denote the surface area of the

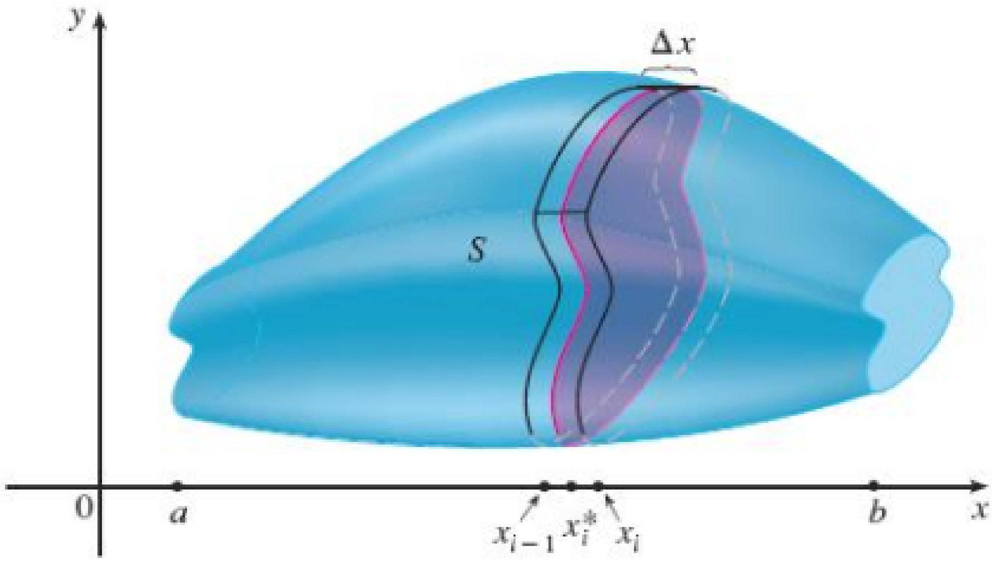


Figure 1.0.4: A slice of the solid

cross section of the slice. The height of the slice is $\Delta x = x_i - x_{i-1}$. Then

$$\text{the volume of one such slice} = A(x_i^*)\Delta x.$$

Let $V(S)$ denote the volume of the entire solid S. Then we have

$$V(S) = \lim_{n\to\infty} \sum_{i=1}^{n} A(x_i^*)\Delta x = \int_a^b A(x)dx,$$

where we assume that $A(x)$ is a continuous function of x.

Remark 1.0.1. Even if the given solid is not shaped in the above-mentioned way, we can make it like this through some mechanism. We will see later on that we will be able to find out a way even then.

Example 1.0.2. Consider a sphere of radius r centered at the origin O. Let $A(x)$ denote the area of the cross section when the sphere is cut by the plane P_x. Clearly then,

$$A(x) = \pi y^2,$$

where y is the radius of the cross sectional disk. Please see Figure 1.0.5 for an illustration. Therefore,

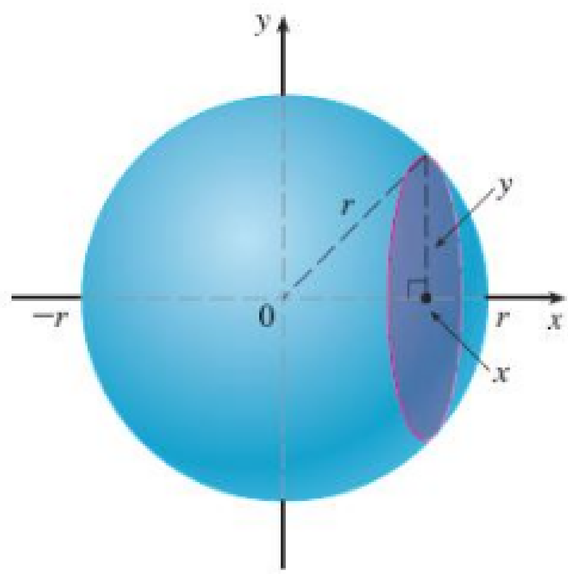


Figure 1.0.5: A sphere of radius r

$$A(x) = \pi y^2 = \pi(r^2 - x^2).$$

Then the volume V of the sphere is given by

$$V = \int_{-r}^{r} \pi(r^2 - x^2)dx = \pi \left[r^2 x - \frac{x^3}{3} \right]_{-r}^{r} = \frac{4}{3}\pi r^3.$$

□

Example 1.0.3. Find the volume of the solid obtained by rotating the region bounded by $y = x^3, y = 8, x = 0$ about the y-axis. See Figure 1.0.6 below. We will have to consider a disk as shown in red color in part (b) of Figure 1.0.6. The radius of this cross sectional disk is going to be x. Then the volume V of the solid is given by

$$V = \int_{0}^{8} (\pi x^2)dy$$

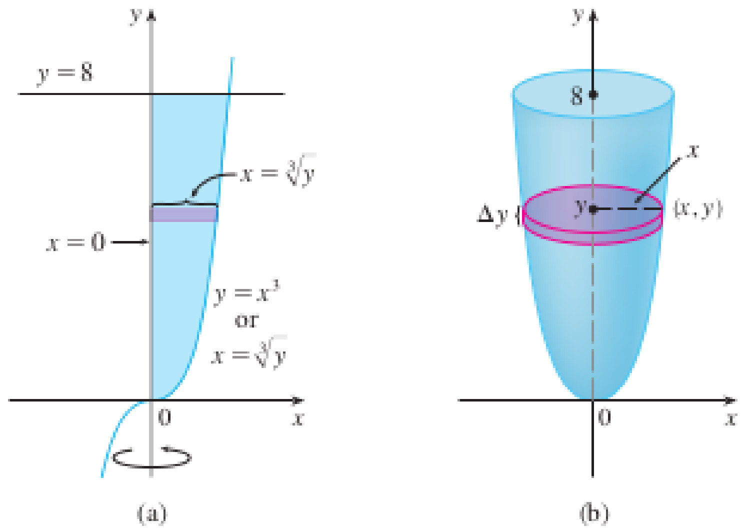


Figure 1.0.6: Figure for Example 1.0.3

As $x = y^{1/3}$, we have:

$$V = \int_0^8 (\pi x^2) dy = \int_0^8 \pi y^{2/3} dy.$$

The rest of the calculation is left to you as an exercise. □

The above example has given you some idea about how to find out the volume of a solid of revolution. In the above example, we had a single graph of a function of one variable, which was generating the solid. But under certain circumstances, the solid generated by the revolution may not be like this. The following example discusses one such case.

Example 1.0.4. Let R be the region bounded by the line $y = x$ and the curve $y = x^2$. Find the volume of the solid of revolution when the region R is rotated about the x-axis.

Observe that if we cut the solid (generated by the rotation) by a plane perpendicular to the x-axis, we are going to get a shape like a "washer" (as shown in part (c) of Figure 1.0.7).

Let y_2 denote the distance of the curve $y = x^2$ from the x-axis and y_1 denote

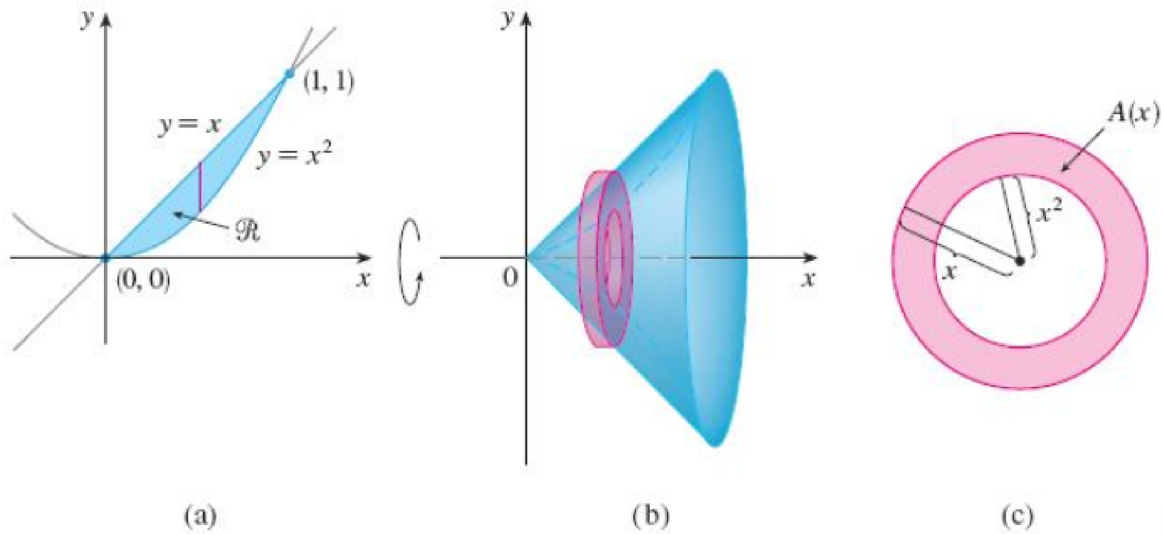


Figure 1.0.7: Figure for Example 1.0.4

the distance of the line $y = x$ from the x-axis. Then the area of the shaded portion $A(x)$ (as shown in part (c) of Figure 1.0.7) is given by

$$\pi y_1^2 - \pi y_2^2.$$

And the volume V of the solid is given by

$$V = \int_0^1 \pi(y_1^2 - y_2^2)dx = \int_0^1 \pi[x^2 - (x^2)^2]dx.$$

The rest of the calculation is left to you as an exercise. □

Example 1.0.5. Consider the same region R as in Example 1.0.4 above. Find the volume of the solid of revolution when the region R is rotated about the line $y = 2$. Please see Figure 1.0.8 for an illustration.

If we consider a slice of this solid at a distance x from the origin, we will again get a shape like a "washer" as shown in the right hand side picture in Figure 1.0.8. Let y_1 denote the distance between the line $y = 2$ and the curve $y = x^2$. Clearly then $y_1 = 2 - x^2$. Similarly, if y_2 denotes the distance between the line $y = 2$ and the line $y = x$, then we have $y_2 = 2 - x$. So the

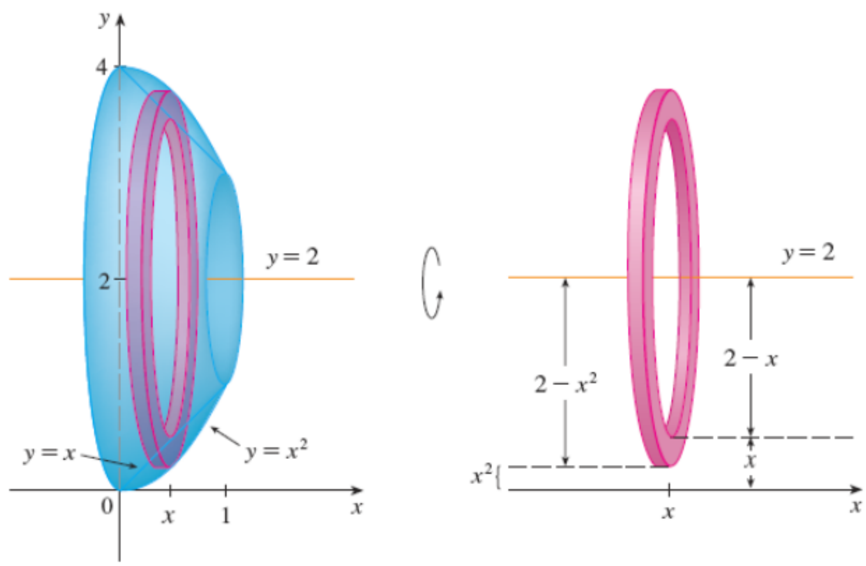


Figure 1.0.8: Figure for Example 1.0.5

volume V of the solid is given by

$$V = \int_0^1 \pi(y_1^2 - y_2^2)dx = \int_0^1 \pi[(2-x^2)^2 - (2-x)^2]dx.$$

The rest of the calculation is left to you as an exercise. $\square$

We are now going to see how to generalize this kind of situation, where the cross section of the solid with a plane P_x (perpendicular to the x-axis), is of the shape of a washer. But before that, we have a small remark.

Remark 1.0.6. Please see Figure 1.0.9 below. As the number of subdivisions increase, the volume of the solid is going to be more and more accurate.

(a) Using 5 disks, $V \approx 4.2726$ (b) Using 10 disks, $V \approx 4.2097$ (c) Using 20 disks, $V \approx 4.1940$

Actual value=4.18879

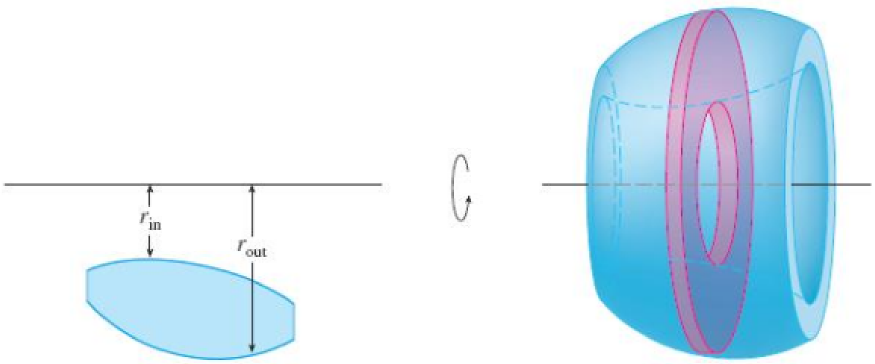
Figure 1.0.9: Figure for Remark 1.0.6

The solids in the Examples are all called **solids of revolution** because they are obtained by revolving a region about a line. In general, we calculate the volume of a solid of revolution by using the basic defining formula

$$V = \int_a^b A(x)\,dx \qquad \text{or} \qquad V = \int_c^d A(y)\,dy$$

and we find the cross-sectional area $A(x)$ or $A(y)$ in one of the following ways:

- If the cross-section is a disk , we find the radius of the disk (in terms of x or y) and use
$$A = \pi(\text{radius})^2$$
- If the cross-section is a washer , we find the inner radius r_{in} and outer radius r_{out} from a sketch and compute the area of the washer by subtracting the area of the inner disk from the area of the outer disk:
$$A = \pi(\text{outer radius})^2 - \pi(\text{inner radius})^2$$



2 Multiple integrals

In this topic, we are going to deal with problems of the type given in the 2 examples below.

Example 2.0.1. Suppose we consider the function $f(x, y) = 16 - x^2 - 2y^2$. Find the volume below the surface $z = f(x, y)$ which lies above the xy-plane. Please refer to Figure 2.0.1. $\square$

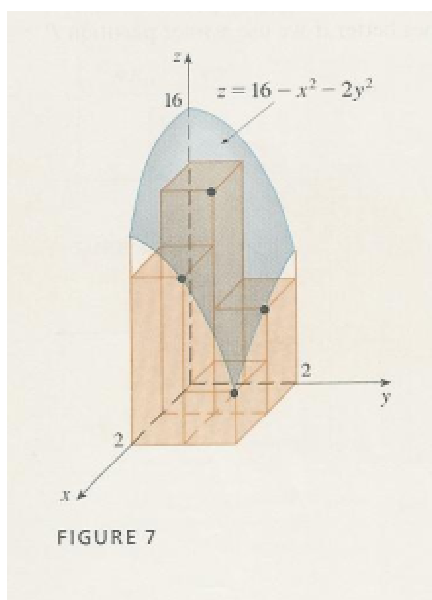


Figure 2.0.1: Figure for Example 2.0.1

Example 2.0.2. Find the volume of the solid which lies above the xy-plane, below the paraboloid $z = x^2 + y^2$, and between the plane $y = 2x$ and the parabolic cylinder $y = x^2$. Please refer to Figure 2.0.2 for this.

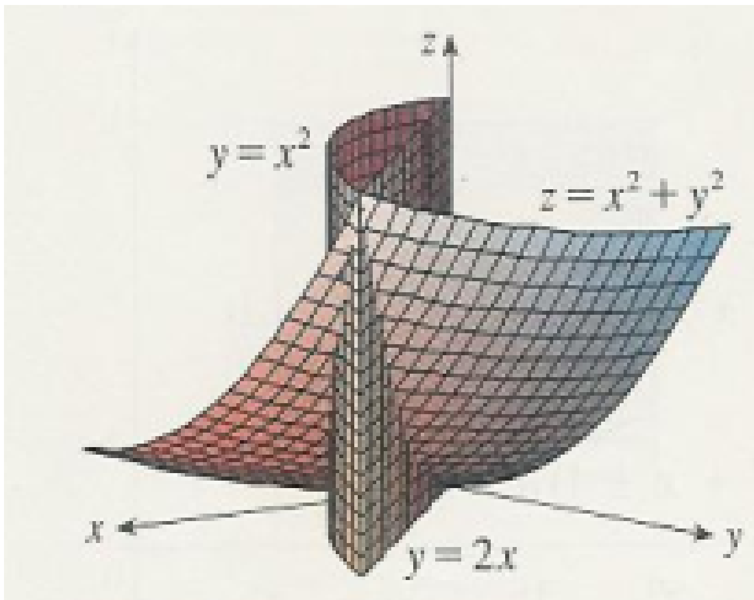


Figure 2.0.2: Figure for Example 2.0.2

□

2.1 Review of definite integral for a function of a single variable

Suppose we are considering a continuous function $y = f(x)$ on the interval $[a, b]$. In order to find out the definite integral of $f(x)$ between the points a and b, we divided the interval $[a, b]$ into certain subintervals $[x_{i-1}, x_i]$, and on each subinterval, we considered a rectangle whose breadth was the length of the subinterval, and the height was nothing but $f(x_i^*)$, where x_i^* was any point in the subinterval $[x_{i-1}, x_i]$. Please refer to Figure 2.1.1 for this.

And then we took the sum $\sum_{i=1}^{n} f(x_i^*)\Delta x$ over all such rectangles and then took limit as $n \to \infty$. That gave us the definite integral as

$$\int_a^b f(x)dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x.$$

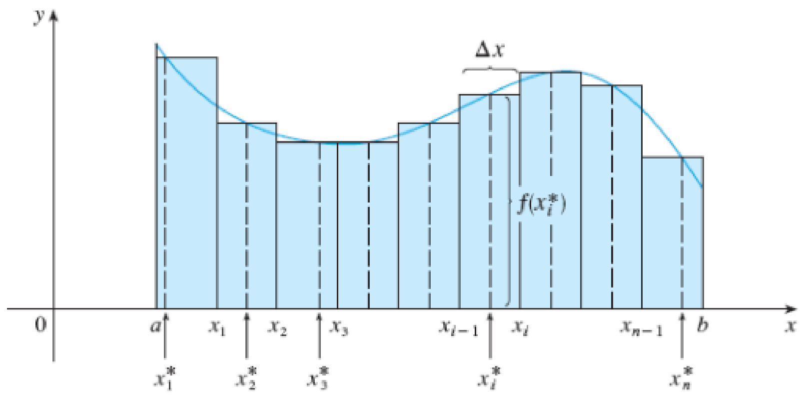


Figure 2.1.1: Review of definite integral

2.2 Double integrals over rectangles

Consider a continuous function $f(x, y)$ of 2 variables over the rectangle

$$R = \{(x, y) \in \mathbb{R}^2 | a \leq x \leq b, c \leq y \leq d\}.$$

Assume that $f(x, y) \geq 0$ for all $(x, y) \in R$. That is, the graph of the function f will lie above the xy-plane. Please see Figure 2.2.1 for an illustration.

Let us divide the interval $[a, b]$ into n many subintervals $[x_{i-1}, x_i]$ and $[c, d]$

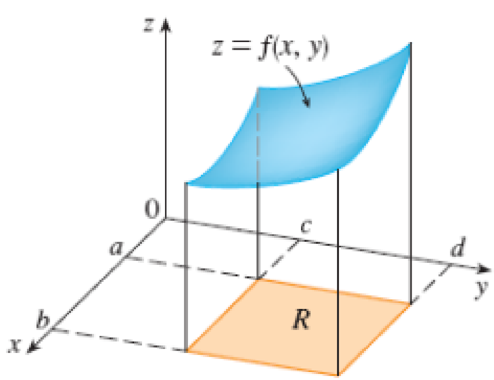


Figure 2.2.1: A function of 2 variables over a rectangle R

into m many subintervals $[y_{j-1}, y_j]$ (The subintervals need not be of equal length). Consider the sub-rectangle $R_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j]$ and consider a point (x_{ij}^*, y_{ij}^*) in this sub-rectangle. Please refer to Figure 2.2.2 for an illustration.

Then corresponding to each such sub-rectangle R_{ij}, we will get a rectangular

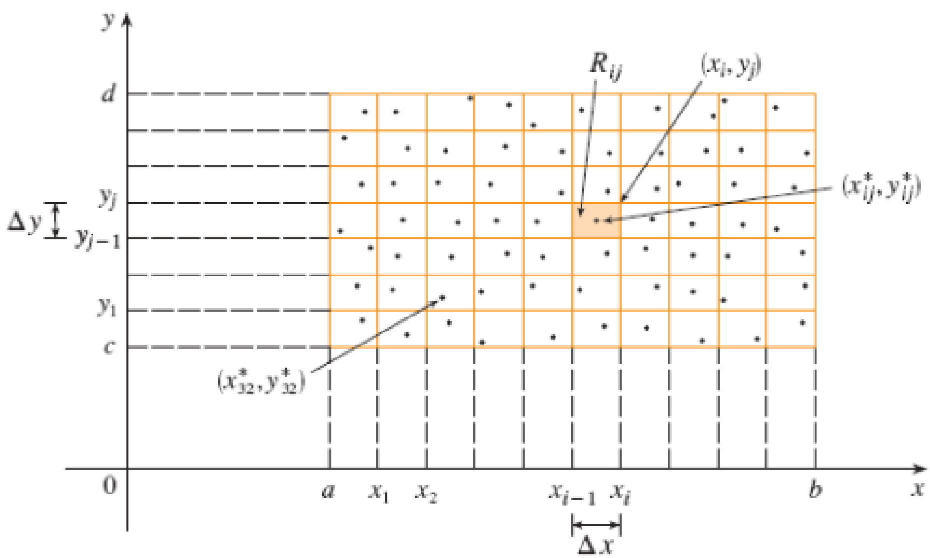


Figure 2.2.2: The sub-rectangles R_{ij} of the rectangle R

box whose height is given by $f(x_{ij}^*, y_{ij}^*)$. Please see Figure 2.2.3 for this. Now if we multiply $f(x_{ij}^*, y_{ij}^*)$ by the area ΔA_{ij} of the sub-rectangle R_{ij}, then we will get the volume V_{ij} of the small rectangular box in Figure 2.2.3 as

$$V_{ij} = f(x_{ij}^*, y_{ij}^*)\Delta A_{ij},$$

where $\Delta A_{ij} = \Delta x_i \times \Delta y_j$. If we consider the volume of all such rectangular boxes (over all such sub-rectangles), and take their sum, then the sum is going to be

$$\sum_{i=1}^{n} \sum_{j=1}^{m} f(x_{ij}^*, y_{ij}^*)\Delta A_{ij}.$$

Figure 2.2.4 shows a collection of the rectangular boxes over all the sub-rectangles.

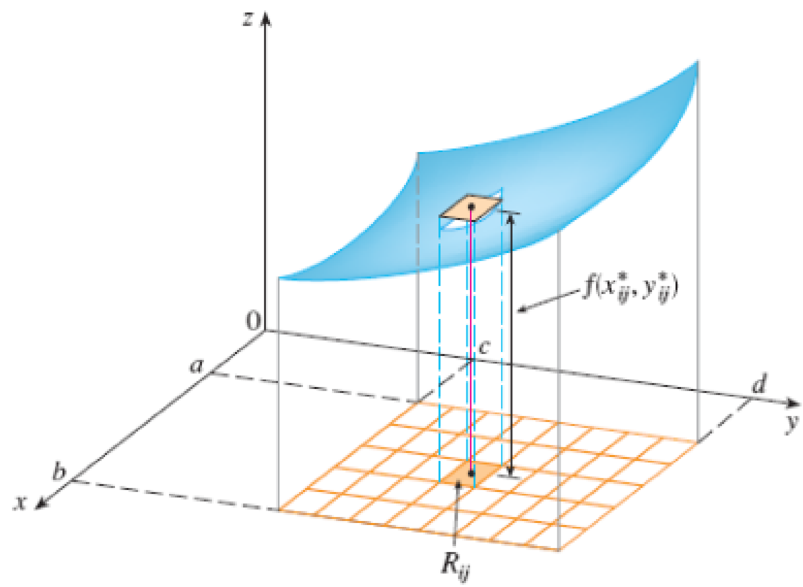


Figure 2.2.3: A rectangular box over the sub-rectangle R_{ij}

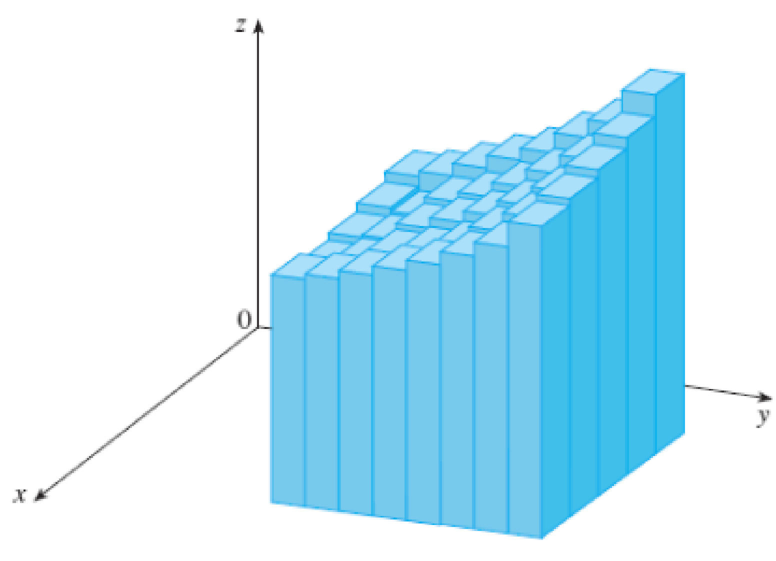


Figure 2.2.4: The rectangular boxes over all sub-rectangles

If we now take limits on this sum as both $m, n \to \infty$, then we will get the volume V of the solid which is above the xy-plane, below the surface $z = f(x, y)$, and bounded by $a \le x \le b, c \le y \le d$. That is,

$$V = \lim_{m,n\to\infty} \sum_{i=1}^{n} \sum_{j=1}^{m} f(x_{ij}^*, y_{ij}^*) \Delta A_{ij}.$$

The expression on the right hand side of the above equation is called the **double integral** of $f(x, y)$ over the rectangle R, and is denoted by $\int_R \int f(x, y) dA$, where $dA = dxdy$.

2.3 Double integrals as volumes

As seen in the previous subsection, we have

$$V = \lim_{m,n\to\infty} \sum_{i=1}^{n} \sum_{j=1}^{m} f(x_{ij}^*, y_{ij}^*) \Delta A.$$

The double integral of f over the rectangle R is

$$\int_R \int f(x, y) dA = \lim_{m,n\to\infty} \sum_{i=1}^{n} \sum_{j=1}^{m} f(x_{ij}^*, y_{ij}^*) \Delta A \qquad (*)$$

provided the limit exists.

Remark 2.3.1. The limit in equation $(*)$ has to exist. Only then we can say that f is integrable over R.

The precise meaning of the limit in equation $(*)$ above is that for every number $\epsilon > 0$, there exists an integer N such that

$$\left| \int_R \int f(x, y) dA - \sum_{i=1}^{n} \sum_{j=1}^{m} f(x_{ij}^*, y_{ij}^*) \Delta A \right| < \epsilon$$

for all integers m, n greater than N, and for any choice of sample points (x_{ij}^*, y_{ij}^*) in R_{ij}.
If we take the sample points (x_{ij}^*, y_{ij}^*) in R_{ij} to be the corner points (x_i, y_j), then equation $(*)$ reduces to equation $(**)$ below:

$$\int_R \int f(x, y) dA = \lim_{m,n\to\infty} \sum_{i=1}^{n} \sum_{j=1}^{m} f(x_i, y_j) \Delta A. \qquad (**)$$

If $f(x, y) \geq 0$, then the volume V of the solid that lies above the rectangle R and below the surface $z = f(x, y)$ is

$$V = \int_R \int f(x, y) dA.$$

Example 2.3.2. Evaluate $\int_R \int \sqrt{1 - x^2} dA$, where $R = \{(x, y) \in \mathbb{R}^2 | -1 \leq x \leq 1, -2 \leq y \leq 2\}$. Please refer to Figure 2.3.1 below.
Let $z = \sqrt{1 - x^2}$. Then $x^2 + z^2 = 1$. This represents a circular disk.

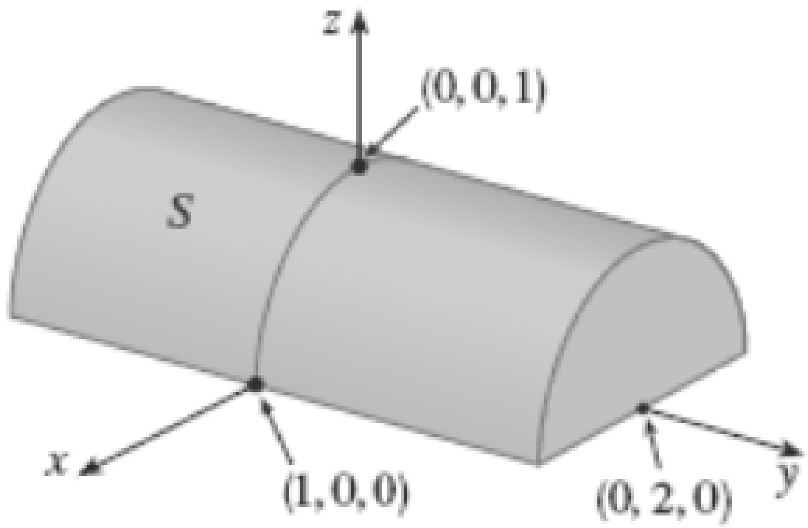


Figure 2.3.1: Figure for Example 2.3.2

The area of the half-circular disk equals $\frac{1}{2}\pi 1^2$. And the height of the solid (half cylinder) as shown in Figure 2.3.1 is 4 (from $y = -2$ to $y = 2$). Therefore, the required double integral value equals

$$\left(\frac{1}{2}\pi 1^2\right) .4 = 2\pi.$$

This is the volume of the half-cylinder shaped solid as shown in Figure 2.3.1.
$\square$

3 Double integrals over rectangles continued...

Remark 3.0.1. Earlier, we had studied the topic "Double integrals over rectangular domains". But in general, the domain over which the double integral is taken is not necessarily a rectangle all the time. We will soon see how to handle such cases, where the domain is not a rectangle.

3.1 Properties of double integrals

If f, g are continuous functions defined over the rectangle R, then

1. $$\int_R\int [f(x,y)+g(x,y)]dA = \int_R\int f(x,y)dA + \int_R\int g(x,y)dA.$$

2. If c is a constant, then
$$\int_R\int cf(x,y)dA = c\int_R\int f(x,y)dA.$$

3. If $f(x,y) \geq g(x,y)$ for all $(x,y) \in R$, then
$$\int_R\int f(x,y)dA \geq \int_R\int g(x,y)dA.$$

Here dA is nothing but $dxdy$ or $dydx$. The above statements are left to the reader as an **exercise**.

4 Iterated integrals

Till now, we have not talked about the order in which we are going to take the limits $a \leq x \leq b$ and $c \leq y \leq d$ of integration. We will study this in this section.

4.1 Partial integration

We know that

$$\int_R\int f(x,y)dA = \int_{x=a}^{b}\int_{y=c}^{d} f(x,y)dydx.$$

Suppose we consider $\int_{y=c}^{d} f(x,y)dy$. We know that f is a function of 2 variables x and y. In the above integral $\int_{y=c}^{d} f(x,y)dy$, if we assume that x is constant, and integrate with respect to y, then we are eventually going to get a function of x, say $A(x)$. That is,

$$\int_{y=c}^{d} f(x,y)dy = A(x).$$

This procedure is known as **partial integration**. Now we have

$$\int_R\int f(x,y)dA = \int_{x=a}^{b}\left[\int_{y=c}^{d} f(x,y)dy\right]dx = \int_{x=a}^{b} A(x)dx.$$

The integral $\int_{x=a}^{b} A(x)dx$ is going to give us a number, which is the value of the double integral.
Here, we had kept x fixed first, and then integrated partially with respect to y. Now the same thing can be done by keeping y fixed first, and then integrating the integrand partially with respect to x. Then we would have obtained

$$\int_R\int f(x,y)dA = \int_{y=c}^{d}\left[\int_{x=a}^{b} f(x,y)dx\right]dy = \int_{y=c}^{d} B(y)dy,$$

where $B(y) = \int_{x=a}^{b} f(x,y)dx$.

4.2 Fubini's theorem

Definition 4.2.1. The integrals $\int_{y=c}^{d}\left[\int_{x=a}^{b} f(x,y)dx\right]dy$ and $\int_{x=a}^{b}\left[\int_{y=c}^{d} f(x,y)dy\right]dx$ are called **iterated integrals**. □

Theorem 4.2.2. ***Fubini's theorem:*** *If f is continuous on the rectangle*

$$R = \{(x,y)|a \leq x \leq b, c \leq y \leq d\},$$

then

$$\int_R \int f(x,y)dA = \int_a^b \int_c^d f(x,y)dydx = \int_c^d \int_a^b f(x,y)dxdy.$$

More generally, this is true if we assume that f is bounded on R, f is discontinuous only on a finite number of smooth curves, and the iterated integrals exist.

Remark 4.2.3. Fubini's theorem tells us that if f is continuous on R, then the order of integration can be interchanged. The final value of the integral is independent of this order.

Figure 4.2.1 illustrates Fubuni's theorem when x is fixed. Suppose we keep x fixed, then integrating partially with respect to y will give us a function of x, say $A(x)$, and this function $A(x)$ can be thought of as the area of the red colored region in Figure 4.2.1. This area is nothing but a cross section of the solid exactly at the location x. In the last lecture, we had talked about the volume integral. For getting the volume here, we have to consider such cross sections and then take the sum (or integration) over all such cross sections.

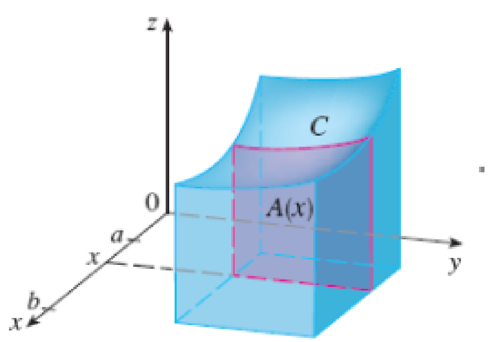


Figure 4.2.1: Fubini's theorem (when x is fixed)

One can provide a similar illustration of Fubini's theorem when y is fixed. See Figure 4.2.2 below.

Example 4.2.4. Evaluate $\int_R \int y\sin(xy)dA$, where $R = [0,2] \times [o, \pi]$. Now the question is: Should we integrate first with respect to x or y?

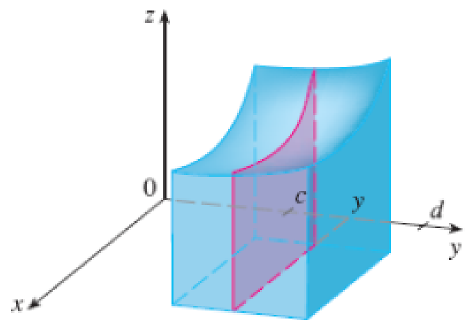


Figure 4.2.2: Fubini's theorem (when y is fixed)

By Fubini's theorem, it is going to work both ways. But we will have to see which strategy will give us the smoothest procedure.
For this example, observe that if we integrate with respect to x first, then we are going to get a function of y. But if we integrate first with respect to y, then it is going to be more difficult, because the integrand is a product of 2 functions of y, and we will have to do integration by parts in that case. That's why, we will integrate with respect to x first. Then the given integral equals

$$= \int_{y=0}^{\pi} \left[\int_{x=0}^{2} y \sin(xy) dx \right] dy = \int_{y=0}^{\pi} \left[\frac{y(-\cos(xy))}{y} \right]_0^2 dy$$

$$= -\int_0^{\pi} [\cos(2y) - 1] dy.$$

The rest of the calculation is left to you as an exercise. □

5 Double integrals over general regions

Recall that the domain of definition of a function need not have to be a rectangle in the 2 dimensional plane. It can be arbitrarily shaped. Let D be the domain of a continuous function $f(x, y)$. We will assume that D is bounded. That means, we will be able to enclose the region D inside a finite rectangle R. The rectangle doesn't have to be such that it touches the boundaries of D. It can be any finite rectangle. Please see Figure 5.0.1 for an illustration.

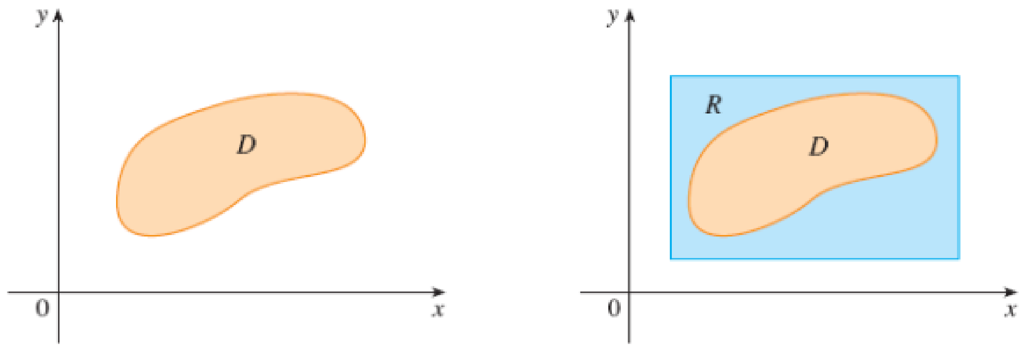


Figure 5.0.1: The region D enclosed in a rectangle R

Recall that our aim was to find $\int_D \int f(x,y)dA$. Define a new function $F(x,y)$ as follows:

$$F(x,y) = \begin{cases} f(x,y) \text{ if } (x,y) \in D \\ 0 \text{ if } (x,y) \in R \setminus D \end{cases}$$

Clearly then, F is going to be discontinuous on the boundary of D. But we will have no problem with it if the boundary curve of D is well defined. The graph of $f(x,y)$ will look as in Figure 5.0.2 below.

The graph of $F(x,y)$ will look like the blue colored portion in Figure 5.0.3 below.

If F is integrable over R, then we define the double integral of f over D by

$$\int_D \int f(x,y)dA = \int_R \int F(x,y)dA.$$

Justification: Since $R = D \cup D^c$, we have

$$\int_R \int F(x,y)dA = \int_D \int F(x,y)dA + \int_{D^c} \int F(x,y)dA$$

$$= \int_D \int f(x,y)dA + \int_{D^c} \int 0dA$$

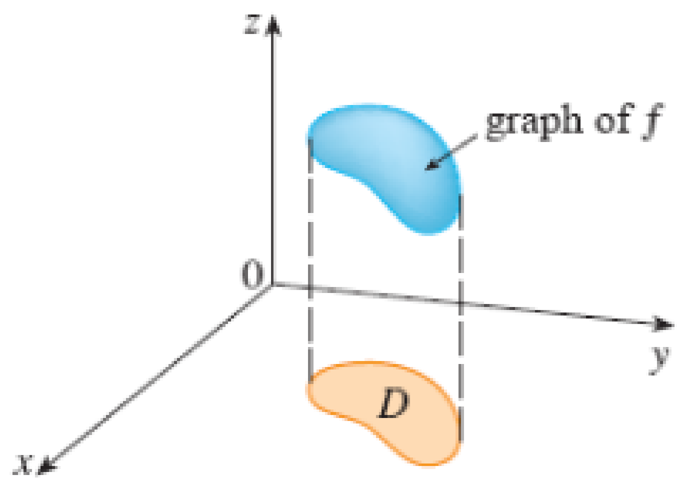


Figure 5.0.2: The graph of $f(x, y)$

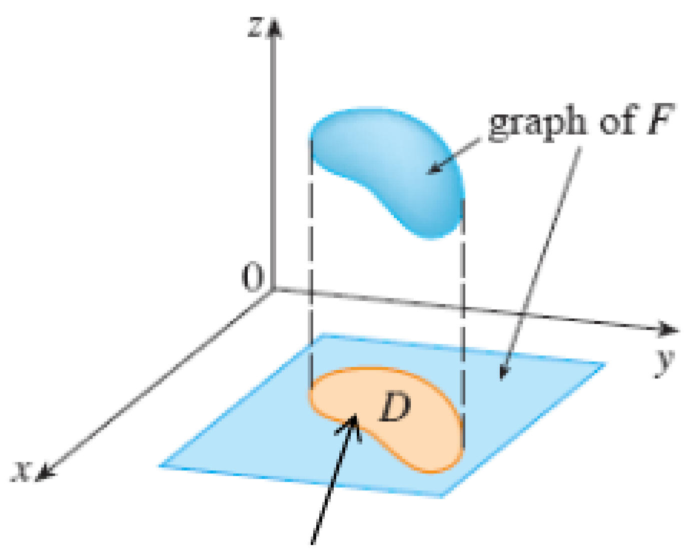


Figure 5.0.3: The graph of $F(x, y)$

$$= \int_D \int f(x,y)dA + 0 = \int_D \int f(x,y)dA.$$

Whenever we are given to find out a double integral of a function over a general region D, the first task is to define the general region D and to enclose it inside a finite rectangular domain.

5.1 Type I and II regions

We are now going to categorize any general region D as either "type I or type II region". Then we will see that any general region can be expressed as a union of type I region or type II regions or both.

Definition 5.1.1. A region D is said to be of **type I** if it lies between 2 continuous curves defined as functions of x. That is,

$$D = \{(x,y)|a \le x \le b, g_1(x) \le y \le g_2(x)\}.$$

□

Figure 5.1.1 illustrates the concept of a type I region.

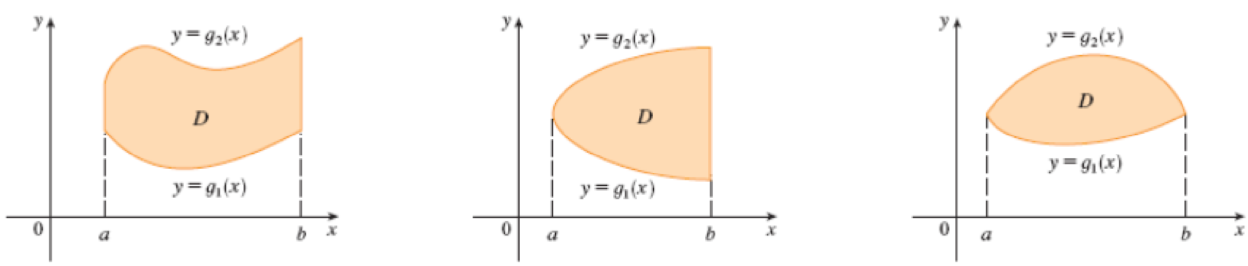


Figure 5.1.1: Some type I regions

If we want to convert $\int_D \int f(x,y)dA$ (where D is a type I region) into $\int_R \int F(x,y)dA$, then we can easily get some rectangle R in which D can be enclosed. Please see Figure 5.1.2 for this.

When $g_1(x) \le y \le g_2(x)$, then definitely $F(x,y) = f(x,y)$. But if $y \le g_1(x)$

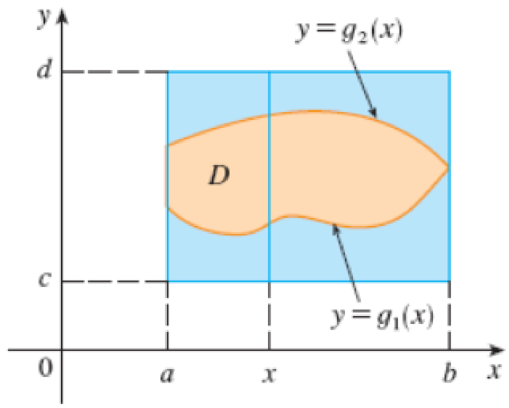


Figure 5.1.2: A type I region enclosed in a rectangle

or $y \geq g_2(x)$, then $F(x, y) = 0$. If f is continuous on a type I region D such that

$$D = \{(x, y)|a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\},$$

then

$$\int_D \int f(x, y)dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)dydx.$$

Similarly, we can consider **type II** regions, which can be expressed as

$$D = \{(x, y)|c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\},$$

where h_1 and h_2 are continuous. Figure 5.1.3 illustrates the concept of a type II region. For a type II region D, we have:

$$\int_D \int f(x, y)dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y)dxdy.$$

Sometimes, we may be able to express a given region both as type I and type II (We will see examples for this later on.). In that case, we will have to choose that kind of region for which the entire procedure of integration will be simpler.

Example 5.1.2. Evaluate $\int_D \int (x + 2y)dA$, where D is the region bounded by the parabolas $y = 2x^2$ and $y = 1 + x^2$.

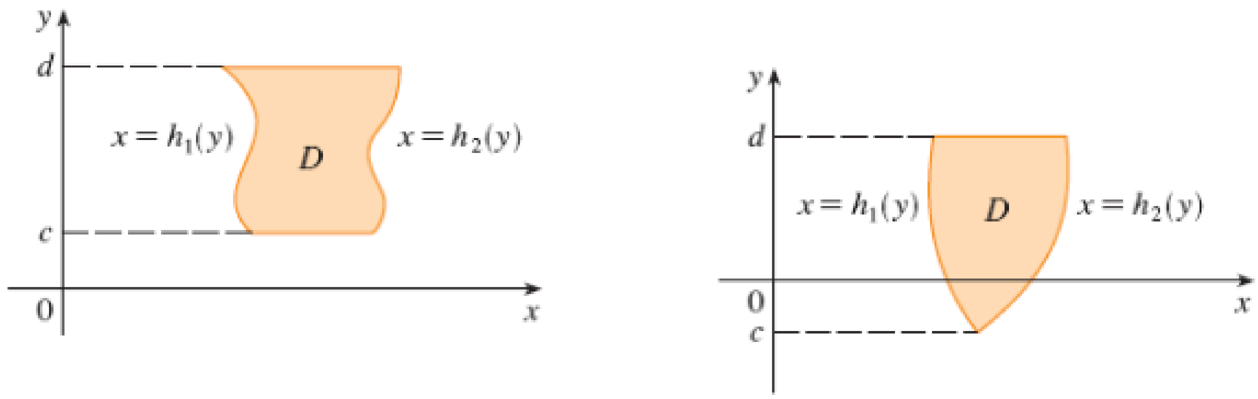


Figure 5.1.3: Some type II regions

Firstly, we will have to find out the domain D. The points of intersection of the 2 curves $y = 2x^2$ and $y = 1 + x^2$ are $(1, 2)$ and $(-1, 2)$. Clearly then

$$D = \{(x, y)| - 1 \leq x \leq 1, 2x^2 \leq y \leq 1 + x^2\}.$$

Figure 5.1.4 depicts the region D. Here D is a type I region.

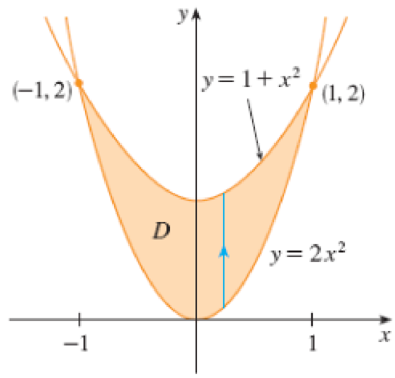


Figure 5.1.4: Figure for Example 5.1.2

Then

$$\int_D\int (x+2y)dA = \int_{x=-1}^{1}\int_{y=2x^2}^{1+x^2}(x+2y)dydx$$

$$= \int_{-1}^{1}\left[xy+y^2\right]_{2x^2}^{1+x^2}dx = \int_{-1}^{1}[x(1+x^2-2x^2)+(1+x^2)^2-4x^4]dx.$$

The rest of the calculation is left to the reader as an exercise. □

Remark 5.1.3. Whenever we consider D as a type I region, the limits of the y variable are going to be functions of x, and we will have to consider the bottom and the top functions. Similarly, for type II regions, we will have to consider the left and the right functions. It is all about finding the domain D.

Example 5.1.4. Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$ and above the region D in the xy-plane which is bounded by the line $y = 2x$ and the parabola $y = x^2$. Figure 5.1.5 depicts the solid.

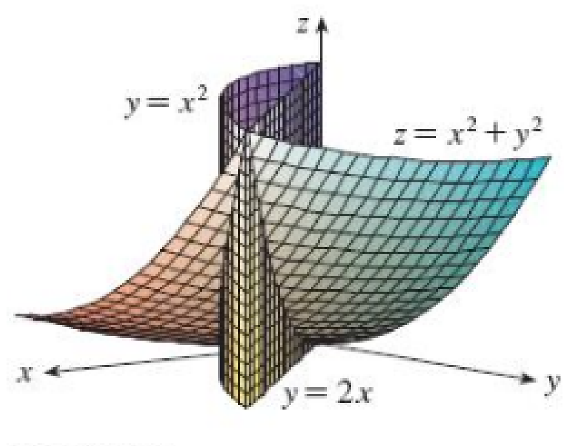


Figure 5.1.5: Figure for Example 5.1.4

Basically, this volume is given by $\int_D\int(x^2+y^2)dA$, where D is the region in the xy-plane bounded by $y = 2x$ and $y = x^2$.
The points of intersection of $y = 2x$ and $y = x^2$ are $(0, 0)$ and $(2, 4)$. Observe that this region D can be expressed both as type I and type II regions. As a type I region,

$$D = \{(x, y)|0 \le x \le 2|x^2 \le y \le 2x\}.$$

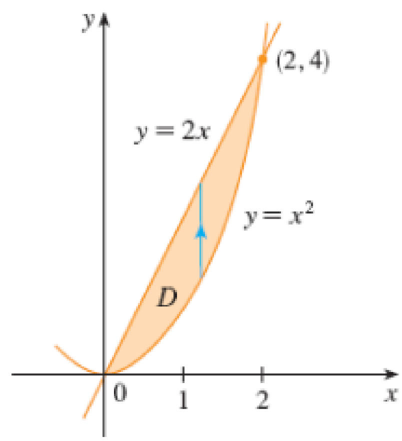


Figure 5.1.6: D as a type I region (Example 5.1.4)

Figure 5.1.6 depicts D as a type I region.

And as a type II region,

$$D = \{(x, y) | 0 \leq y \leq 4, \frac{y}{2} \leq x \leq \sqrt{y}\}.$$

Figure 5.1.7 depicts D as a type II region.

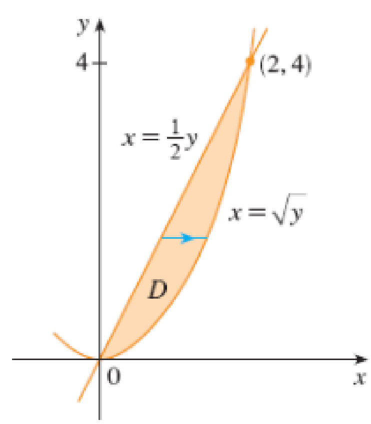


Figure 5.1.7: D as a type II region (Example 5.1.4)

The details of the calculation are left to the reader as an exercise. Please check that both the approaches give the same answer. □

————————-END————————